

Slaughterhouse documentation

Contents

Slaughterhouse documentation	1
1. Subsystem Use Cases	2
A. UC-01: Receiving Delivery Notice from Management	2
B. UC-02: Create a Slaughter Lot.....	3
C. UC-03: Create Ante-Mortem Inspection Sheet.....	4
D. UC-04: Monitor Daily Stats	5
E. UC-05: Approve the Inspection Sheet.....	6
F. UC-06: Create Delivery Notice	7
G. UC-07: Processing Starts	8
H. UC-08: Processing finishes	9
I. UC-09: Update Cold Storage Log.....	10
J. UC-10: Create Waste Report	10
K. UC-11: Create Slaughter Report.....	11
2. Use Case Diagram	12

1. Subsystem Use Cases

A. UC-01: Receiving Delivery Notice from Management

Use Case ID: UC-01

Use Case Name: Receiving Delivery Notice from Management

Actor: Slaughterhouse Manager

Preconditions:

- DeliveryNotice must be sent before this scenario will happen.
- Scheduled delivery date/time set.
- System integration with Slaughterhouse active.

Main Flow:

- The Manager will confirm the receipt of this note from the Management subsystem
- System receives the following data package from the Management.
 - Delivery notice ID
 - Scheduled arrival date/time
 - Lot information (ID, age, breed)
 - Estimated quantity and weight
 - Average bird weight
 - Quality grade
 - Health status
 - Treatment history summary
 - Withdrawal period confirmations
 - Special handling instructions
 - Transport details
 - Contact information
- System updates DeliveryNotice status to Received
- System records receipt timestamp

Alternative Flows:

- **A1: Missing Critical Data**
 - Returns to Management System
 - System lists missing fields
- **A2: Slaughterhouse Rejects Delivery**
 - System receives rejection message with reason
 - System notifies Farm Manager
 - System suggests alternative dates
 - Farm Manager must reschedule

Postconditions:

- DeliveryNotice received successfully
- Status updated to RECEIVED
- SlaughterLot will be created and linked to this DeliveryNotice

Business Rules:

- All Notices must be send at least 24 hours before scheduled delivery
- After receiving the Notice, the Slaughterhouse manager must create the Slaughter Lot document

Integration Points:

- **Inbound from Management:**
 - Delivery schedule
 - Lot details
 - Quality metrics
 - Health history
 - Treatment records

B. UC-02: Create a Slaughter Lot

Use Case ID: UC-02

Use Case Name: Create a Slaughter Lot

Actor: Slaughterhouse Manager

Preconditions:

- A valid **Delivery Notice** must be received and marked as Received.
- Chickens must be physically delivered to the Slaughterhouse.

- The Delivery Notice must include lot details (ID, breed, age, weight, etc.).

Main Flow:

- The Manager accesses the Create Slaughter Lot module.
- The system retrieves data from the associated Delivery Notice.
- The Manager reviews and confirms the following data:
 1. Delivery Notice ID
 2. Supplier farm and transport details
 3. Arrival timestamp and lot identifiers
- Quantity and total weight of chickens
 - The Manager validates or adjusts the values if needed.
 - The system creates a new **Slaughter Lot**, automatically linking it to the Delivery Notice.
 - System updates the Lot status to Ready for Inspection.
 - Timestamp of creation is recorded.

Alternative Flows:

A1: Missing Data:

- System notifies the user of incomplete fields.
- Manager must correct data before confirming.

A2: Delivery Rejected:

- Manager can cancel the creation.
- System records reason for cancellation.

Postconditions:

- A new Slaughter Lot is created and linked to its Delivery Notice.
- The lot is now available for Ante-Mortem Inspection.

Business Rules:

- Each Delivery Notice can generate only one Slaughter Lot.
- A Slaughter Lot cannot proceed to processing without completed inspection.

C. UC-03: Create Ante-Mortem Inspection Sheet

Use Case ID: UC-03

Use Case Name: Create Ante-Mortem Inspection Sheet

Actor: Slaughterhouse Manager

Preconditions:

- A Slaughter Lot must exist and be marked as Ready for Inspection.
- The Vet must be logged in with proper access rights.

Main Flow:

- The Vet opens the Ante-Mortem Inspection Sheet creation form.
- The system retrieves the linked Slaughter Lot data.
- The Vet enters or confirms the following:
 - Lot ID, date, and time of inspection
 - Visual condition of birds
 - Mortality rate upon arrival
 - General health observations
 - Temperature, behavior, and physical anomalies
 - The Vet saves the Inspection Sheet as Draft or Pending Approval.
 - System stores the sheet and links it to the respective lot.

Alternative Flows:**A1: Missing or inconsistent data:**

- System displays validation errors.
- Vet corrects and resubmits.

A2: Vet unavailable:

- System allows Quality Control Officer to perform inspection temporarily.

Postconditions:

- Inspection Sheet created and linked to the Slaughter Lot
- Lot status updated to Awaiting Approval.

Business Rules:

- Ante-Mortem inspection must be completed before processing starts.
- Only authorized veterinary or QC staff can create the inspection sheet.

D. UC-04: Monitor Daily Stats

Use Case ID: UC-04

Use Case Name: Monitor Daily Stats

Actor: Slaughterhouse Manager

Preconditions:

- Daily processing operations must be recorded in the system.
- All related sheets (Inspection, Slaughter, Packaging, Waste) must exist for that day.
- The Manager must have read-only or administrative access.

Main Flow:

- The Manager accesses the Daily Statistics dashboard.
- The system aggregates data from all relevant modules:
 - Total birds received and processed
 - Total carcasses rejected
 - Average yield percentage
 - Total packaged and delivered quantities
 - Waste and loss metrics
 - The Manager can apply filters (by date, lot, or production line).
 - The system displays charts and summaries for quick analysis.
 - Manager exports or prints daily reports as needed.

Alternative Flows:**A1: Missing data:**

- System notifies the Manager that some sheets are incomplete.
- Partial statistics are shown with a warning.

Postconditions:

- Daily operational metrics successfully displayed and optionally exported.

Business Rules:

- Daily stats must always match totals in Slaughter and Waste reports.
- Only Managers and authorized users can access cumulative data.

E. UC-05: Approve the Inspection Sheet

Use Case ID: UC-05

Use Case Name: Approve the Inspection Sheet

Actor: Veteranian

Preconditions:

- The Ante-Mortem Inspection Sheet must exist in Pending Approval state.
- The Vet or QC Officer must have approval rights.

Main Flow:

- The Vet reviews the completed inspection data.

- The system displays all relevant fields: health status, mortality, anomalies, etc.
- The Vet clicks “Approve” to confirm the inspection.
- System updates the sheet status to Approved.
- The linked Slaughter Lot status changes to Ready for Processing.
- Timestamp and approver credentials are recorded.

Alternative Flows:

A1: Rejection:

- Vet clicks “Reject”.
- System requests reason for rejection.
- Status changes to Rejected, and the Manager is notified.

A2: Missing data:

- System blocks approval and highlights missing or invalid fields.

Postconditions:

- Inspection sheet status updated to Approved or Rejected.
- Slaughter Lot proceeds or halts based on decision.

Business Rules:

- Only certified veterinary staff may approve inspection sheets.
- Approval is irreversible unless a corrective action process is initiated.

F. UC-06: Create Delivery Notice

Use Case ID: UC-06

Use Case Name: Create Delivery Notice

Actor: Cold Storage Manager

Preconditions:

- Packaged products must exist in Cold Storage and be marked Ready for Dispatch.
- Client order or schedule must be registered in the system.

Main Flow:

- Logistics Manager opens the Create Delivery Notice interface.
- System retrieves available packaged lots.
- Manager selects batches and assigns them to a client order.
- System auto-generates a Delivery Notice with:
 - Notice ID, client details, delivery date
 - Quantity and total weight
 - Product types and packaging details
 - Manager reviews and confirms.
 - System records the Delivery Notice and updates stock accordingly.

Alternative Flows:**A1: Insufficient stock:**

- System alerts the user that requested quantity exceeds available stock.
- Manager must adjust selection.

Postconditions:

- New Delivery Notice created and linked to packaged lots.
- Stock reduced accordingly.

Business Rules:

- Only packaged lots with Approved quality status can be delivered.
- Delivery Notice must reference at least one Packaging Sheet.

G. UC-07: Processing Starts

Use Case ID: UC-07

Use Case Name: Processing Starts

Actor: Workers

Preconditions:

- The Slaughter Lot must have an Approved Ante-Mortem Inspection Sheet.
- Production line and equipment must be ready and operational.

Main Flow:

- Supervisor opens the Processing Control Panel.
- System verifies all preconditions (lot approval, equipment readiness).
- Supervisor selects Start Processing.
- System changes the Slaughter Lot status to In Progress.
- Time of start is logged automatically.
- The operation data (input birds, line used) begins tracking.

Alternative Flows:**A1: Missing Inspection Approval:**

- System blocks start and displays error

A2: Equipment not ready:

- System delays start until all sensors confirm readiness.

Postconditions:

- Slaughter Lot marked as In Progress.
- Automatic logging of start time and initial parameters initiated.

Business Rules:

- Processing cannot begin without a valid, approved inspection.
 - System must record start events automatically.
-

H. UC-08: Processing finishes

Use Case ID: UC-08

Use Case Name: Processing finishes

Actor: Workers

Preconditions:

- The Slaughter Lot must be in In Progress status.
- Processing operation must have been started and tracked.

Main Flow:

- The Supervisor monitors production through the system dashboard.
- When slaughtering completes, the Supervisor selects **Finish Processing**.
- The system automatically collects operational data:
 - Total number of birds processed
 - Total carcasses rejected
 - Average yield and total meat weight
 - Waste generated
 - System updates the Slaughter Lot status to Processed.
 - A Slaughter Report is automatically generated.
 - System records completion timestamp and user credentials.

Alternative Flows:**A1: System error or incomplete data:**

- System requests manual confirmation of final numbers.
- Supervisor enters missing fields before confirming completion.

Postconditions:

- Slaughter Lot marked as Processed.
- Slaughter Report automatically created and linked to the lot.

Business Rules:

- Processing cannot finish if automatic data collection fails validation.
 - Only Supervisors or authorized Managers can confirm completion
-

I. UC-09: Update Cold Storage Log

Use Case ID: UC-09

Use Case Name: Update Cold Storage Log

Actor: Cold Storage Manager

Preconditions:

- Packaged products or batches must exist and be ready for cold storage entry.
- The Cold Storage module must be active and linked to Packaging Sheets.

Main Flow:

- System automatically logs every new packaged batch entering storage.
- Each log entry includes:
 - Lot ID, Product Type, Packaging ID
 - Entry Date/Time, Quantity, and Temperature
 - When products are shipped, system automatically removes them from inventory.
 - Cold Storage Log is updated in real time.
 - Historical entries remain archived for traceability.

Alternative Flows:

A1: Sensor Failure or Manual Entry Needed:

- System allows authorized users to input data manually.
- Validation required by Manager.

Postconditions:

- Cold Storage Log reflects real-time stock and movements.
- Data synchronized with Delivery Notices and Waste Reports.

Business Rules:

- All stock movements must be time-stamped and traceable.
 - Only the system can update logs automatically. Manual changes require approval.
-

J. UC-10: Create Waste Report

Use Case ID: UC-10

Use Case Name: Create Waste Report

Actor: Workers

Preconditions:

- The Slaughter Report must exist and include data on non-usable parts.
- Processing must be completed.

Main Flow:

- System extracts waste data from the Slaughter Report.
- A new Waste Report is generated with:
 - Lot ID, date, and total processed quantity
 - Weight and type of waste (biological, packaging, etc.)
 - Percentage of total waste vs. input quantity
 - QC Officer reviews and confirms the report.
 - System saves and marks the report as Approved.

Alternative Flows:**A1: Inconsistent Data:**

- QC Officer corrects data manually before approval

A2: System Failure:

- Report creation delayed until automatic sync completes.

Postconditions:

- Waste Report successfully created and linked to the processed lot.
- Waste data available for daily statistics and regulatory reporting.

Business Rules:

- Waste percentage must not exceed regulatory thresholds.
- QC approval mandatory for final report validation.

K. UC-11: Create Slaughter Report

Use Case ID: UC-11

Use Case Name: Create Slaughter Report

Actor: Workers

Preconditions:

- The Slaughter Lot must have status Processed.
- Data collection from processing lines must be complete.

Main Flow:

- System automatically compiles operational data from the processing phase.
- Generated Slaughter Report includes:
 - Total birds processed
 - Total weight before and after processing
 - Rejected carcasses and reasons
 - Average yield percentage

